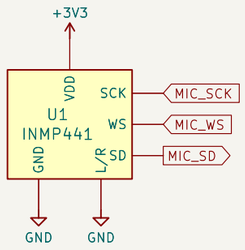
SLOTH CLOCK — WIRING

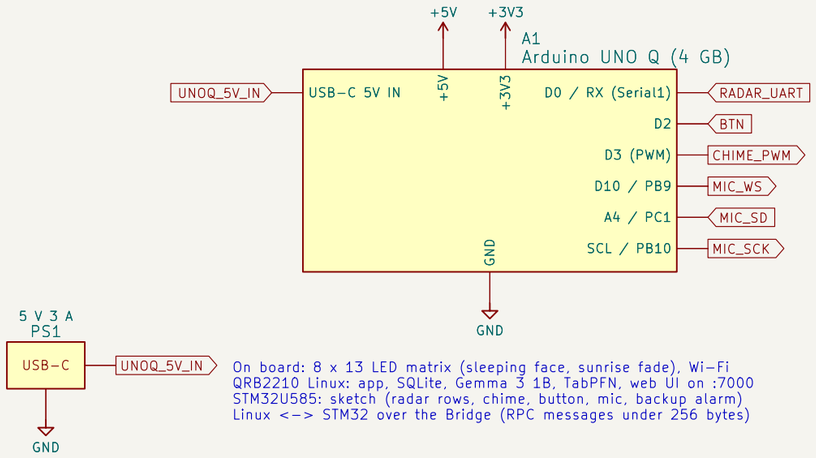
Arduino UNO Q + MR60BHA2 radar kit + INMP441 microphone + PAM8403 chime + EE04 e-paper card

MICROPHONE (INMP441)



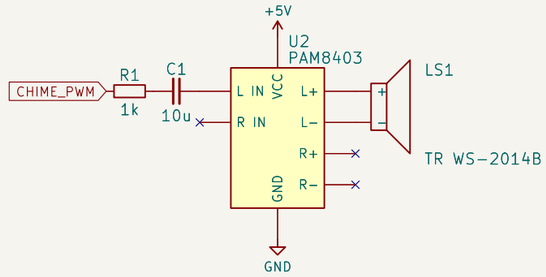
I2S into STM32 SAI1 block A, AF13, register level
L/R to GND = left channel
Header I2C is not usable while the mic runs
Rain only: audio becomes a few numbers per second on the MCU and is thrown away

HOST (Arduino UNO Q)



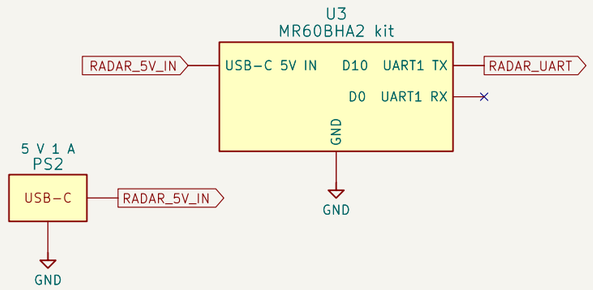
On board: 8 x 13 LED matrix (sleeping face, sunrise fade), Wi-Fi
QRB2210 Linux: app, SQLite, Gemma 3 1B, TabPFN, web UI on :7000
STM32U585: sketch (radar rows, chime, button, mic, backup alarm)
Linux <-> STM32 over the Bridge (RPC messages under 256 bytes)

CHIME (PAM8403 + speaker)



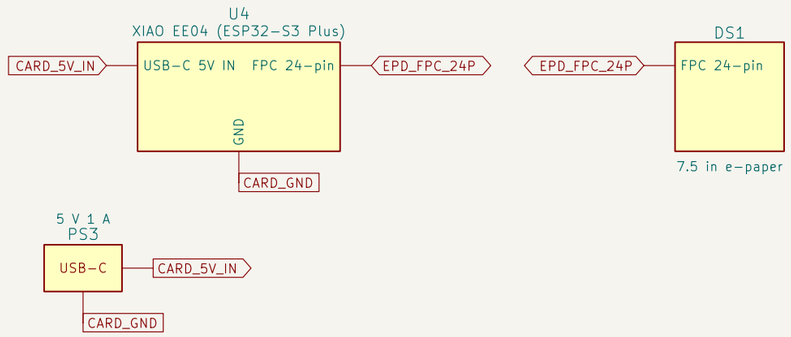
D3 tone() -> 1 k + 10 uF (blocks DC) -> PAM8403 left in
PAM8403 power from the UNO Q 5 V pin
Right channel unused. Keep the volume low.
Soft three-note chime every 30 s, stops after 10 min

RADAR KIT (MR60BHA2 + XIAO ESP32C6)



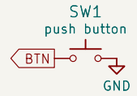
Main link: Wi-Fi, HTTP POST to UNO Q :7000/api/radar
one CSV line per second: R,ms,breath,heart,presence,distance,movement
Wired option: XIAO D10 -> UNO Q D0 (Serial1) plus common GND
Radar talks to the XIAO on UART0 inside the kit; module firmware unchanged
Place 0.5-1.5 m from the chest, pointing at it

E-PAPER CARD (XIAO EE04 + 7.5 in panel)



Wi-Fi only, no wires to the UNO Q. 800 x 480 mono panel (UC8179)
UNO Q -> card: HTTP POST /show (card JSON)
Keys -> UNO Q: HTTP POST :7000/api/card/key
KEY0 (GPIO2) dismiss, hold = step done | KEY1 (GPIO3) next date | KEY2 (GPIO5) refresh
Panel connector jumper on the EE04: 24-pin

BUTTON + PIN MAP



D2 internal pull-up
short = dismiss
hold 2 s = step done

UNO Q PIN	NET	GOES TO
D0 / RX	RADAR_UART	radar kit D10 (wired option)
D2	BTN	SW1 to GND
D3 (PWM)	CHIME_PWM	R1 -> C1 -> PAM8403 L IN
D10 / PB9	MIC_WS	INMP441 WS
A4 / PC1	MIC_SD	INMP441 SD
SCL / PB10	MIC_SCK	INMP441 SCK
+5V	+5V	PAM8403 VCC
+3V3	+3V3	INMP441 VDD
GND	GND	all grounds above
Wi-Fi	—	radar kit, e-paper card

NOTES

All logic is 3.3 V. Each board runs from its own USB 5 V supply: low voltage only, no mains.
Net labels with the same name are connected; no wires are drawn. GND symbols are one common ground.
Privacy: no audio and no raw radar data are stored or sent; only small feature rows go to SQLite on the board.

Parts joined by net labels; radar kit and card talk to the UNO Q over Wi-Fi
UNO Q + MR60BHA2 radar kit + INMP441 + PAM8403 chime + EE04 e-paper card
Sloth Clock | Designed by Abhinav Krishna

Sheet: /
File: sloth_clock.kicad_sch

Title: Sloth Clock — Wiring

Size: A3	Date: 2026-10-26	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 1/1